INTERNSHIP REPORT

On

”Designing a freeform lens free from spherical aberration (Non Physical Model)”

By

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November , 2024

Contents

0.1 Aim

1. Minimizing Spherical Aberration in electrostatic lenses.
2. Constructing geometries of electrodes
3. Collecting potentials and calculating spherical aberration
4. Training the ML model and attempting several geometries for deriving second surface free of aberration.
5. Developing a final model which provides us the appropriate second surface of lens which minimizes the spherical aberration for a given input surface

0.2 Introduction

The above report is a part of a project of achieving an ML model of an Einzel lens with a minimum spherical aberration. The above report mainly focuses on the design of an electrostatic lens system. The mainpurpose of such lens system is to increase its resolution or focusing power and this can be a achieved byreducing the spherical aberration of such lens. Designing lenses to reduce spherical aberration is a pivotalchallenge in optics, for achieving precise imaging in various applications from microscopy to astronomy. In this context, ion optics present an innovative approach both in physical and non-physical models.Addressing spherical aberration, which distorts images by failing to converge light rays to a single focal point, requires laborious engineering and theoretical modeling. Ion optics uses electromagnetic fields to manipulate charged particles, and to electrons or ions, influencing their trajectories to correct aberrationseffectively. This method balances magnetic and electric fields to achieve optimal focusing, thereby minimizing spherical aberration and enhancing image resolution. Practical implementations involve advanced instrumentation and computational simulations to refine the design parameters for optimal performance. It is important to understand the physics of the problem and the implications of the chosen simulation strategy

Artificial neural network model outputs are compared with the SIMION data and very good agreements are found. While artificial neural network is frequently applied in different fields, this is the first study that uses dynamic artificial neural network to predict the parameters of electrostatic lens. The physical model on the other hand explores theoretical frameworks and a general formula to predictand refine lens designs without physical prototypes. Using physical or virtual simulation the common goal of achieving aberration-free imaging through precise lens design.

Ion Optics : Ion optics In the above project we have used ion optics principles that it optics applied to charged particles such as electrons , protons and neutrons .The study of the behavior of ions under the influence of electric and magnetic field is called “ion optics”, because of its similarity to the way light is manipulated through lenses. Light optics is a field that has been studied for more than 400 years, while ion optics is only around for less than 90 years. There are many differences between light optics and ion optics however the main advantage is that they are only convergent. The paraxial ray approximationin electrostatic lenses has been implemented in ion optics that is why here we have focused mostly on three electrode einzel lens system. The Ion Simulation program called Simion is a software to simulate the trajectories of charged particles inside an electrostatic lens system it has a capability of manipulating charged particles under the influence of electric and magnetic fields.

Principles of ion optics:

1. Ion Optics relies on using an Ion lens for focusing In order to focus precisely which can be manipulated by varying potentials across the electrodes.

2. The primary objective is to change the velocity by passing the ions through different electric or magnetic field

3. To change the deflection/trajectories of passed ions which can be done by curve/tilting the potential contours

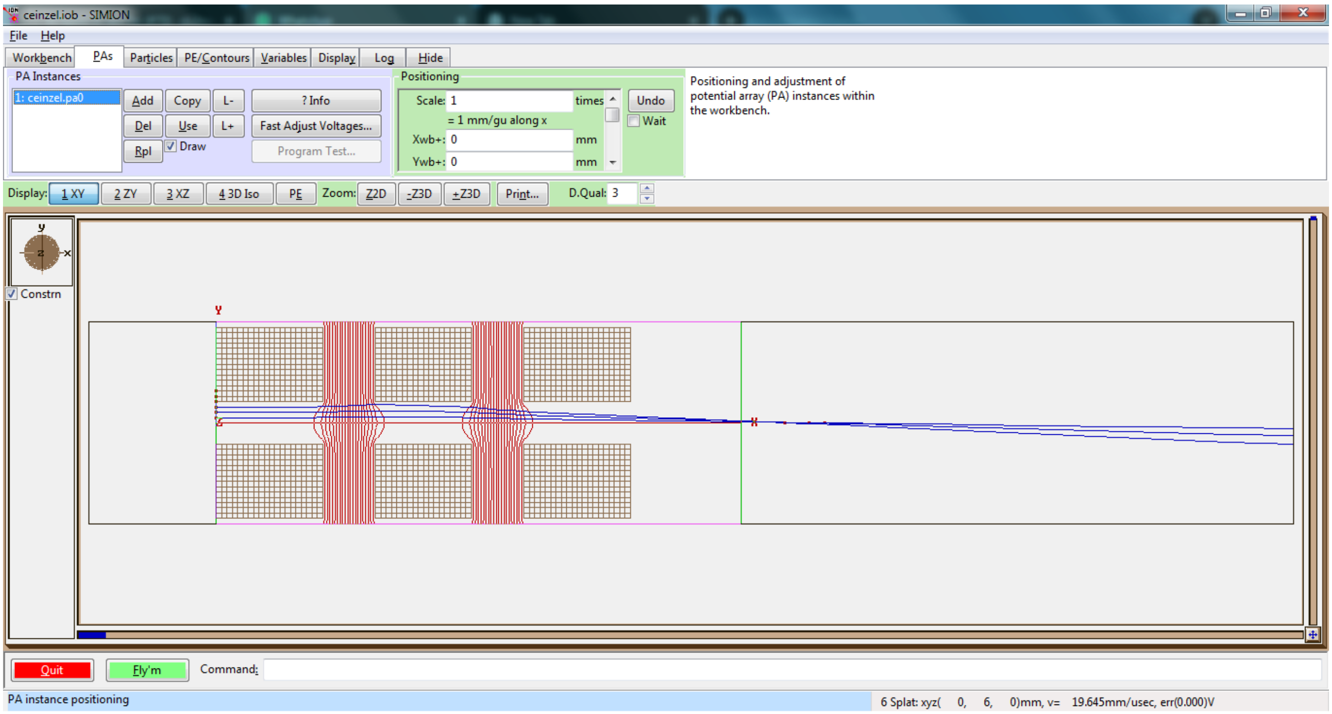
SIMION Software: The Ion simulation program Simion Software is a crucial part of the process we have used it to construct geometries of electrodes such as different types of Einzel lenses and calculating the optical as well as the physical parameters such as velocity, time of flight kinetic energy and most importantly the potential and the deviation of the particles which is nothing but the aberration. SIMION is an electrostatic lens analysis and design program.

In SIMION an electrostatic lens is defined as a two-dimensional electrostatic potential array containing both electrode and non-electrode points. In addition, the user has the option of writing simple programs which can among other actions control field scale factors, dynamically adjust electrodes, and define explicit three-dimensional field functions (e.g. aquadrupole) used in lieu of array fields in specified portions of the potential array. Magnetic fields can be specified for computing ion trajectories in many electrostatic and magnetic field.

- SIMION program originated in Australia in 1973. Don McGilvery created the original SIMION program while working on an undergraduate research project for Professor James Morrison
- The project's goal was to design a double quadrupole mass spectrometer with a common ionization volume for isotope ratio measurements
- A potential array contains a collection of potentials arranged in a square [two-dimensional (2D)] mesh of points.
- The volume represented by a 2D potential array is defined by the array's symmetry assumptions (either planar or cylindrical).

- The potentials of points outside of the electrodes are determined by solving the boundary value problem's Laplace equation via finite difference methods.
- In Applying finite difference methods to a boundary value problem solution (in the simplest form) involves estimating the value of each unknown point by the average of its nearest four (2D) or six (3D) neighboring points. SIMION, this process is called refining the array.
- Successive scans, or iterations, through the array of points gradually converges each point's potential to a stable solution value.
- Finite difference methods use computer memory efficiently
- However, the time to refine an array by finite difference methods is generally proportional to *n^2* where *n* is the number of points in the array.

Interface of SIMION:



Applications: Reducing spherical aberrations in lenses is crucial for achieving high-quality optical performance in various applications:

1. Enhanced Light Transmission: Lenses with reduced spherical aberration transmit more light effectively through the optical system. This is particularly important in applications such as cameras and telescopes, where maximizing the amount of light reaching the sensor or eyepiece improves overall performance and sensitivity.

2. Increased Depth of Field: Spherical aberration can affect depth of field, making it difficult to maintain sharp focus across the entire image plane. Minimizing spherical aberration allows for better control over depth of field, which is beneficial in photography, particularly in portrait and macro photography, where precise focus is critical.

3. Correction in Complex Optical Systems: In advanced optical systems, such as those used in high-power microscopes or telephoto lenses, reducing spherical aberration becomes crucial for maintaining consistent performance across a range of focal lengths and magnifications. This correction ensures that the optical system behaves predictably and reliably under various conditions.

4. Adaptation to Diverse Wavelengths: Different wavelengths of light (colors) can be affected differently by spherical aberration. Reducing aberrations allows lenses to focus various wavelengthsmore accurately, which is important in applications such as spectroscopy and imaging systems thatwork across multiple wavelengths.

5. Advancements in Lens Design: Modern lens design often focuses on minimizing spherical aberration alongside other aberrations like chromatic aberration and coma. Advances in reducing spherical aberrations have led to the development of more compact, lightweight, and higher-performing lenses across different industries, from consumer electronics to scientific research.

0.3 Methodology

In the above project various tools theories and principles has been used .The most important thing to make a model is to first collect the data from a particular geometry upon which the model is going to get trained so the geometry here has been chosen for an Electrostatic lens.

Why Electrostatic lenses? :

1. Electrostatic lenses are convergent never divergent as light as optical lenses often are.

2. Electrostatic lens intersections are always convergent (unlike light optics) forces are always towards the ion optical axis

3. To get divergence in an optical lens one uses a ”cross over” which means one focuses the beam to point after which the trajectories diverge.

A very popular example of such divergence is that is a small spot on a tube in an image of cross over made near the cathode of the electrode gun but those electrostatically weak lenses. A good lens for focusing requires three lens element intersection which can be accelerating -decelerating -accelerating (acel - decal -acel) or decelerating -accelerating decelerating (decel - acel - decel) There mainly two types pf such lenses:

1. Thin lenses

2. Thick lenses

Einzel lens: An Einzel lens is an electrostatic device used for focusing charged particle beams. It maybe found in cathode ray tubes, ion beam and electron beam experiments, as well as in ion propulsion systems. IT consists of three axially aligned cylinders; the outer cylinders are grounded and the cylinder in the middle is held at a fixed voltage as shown below in Figure1

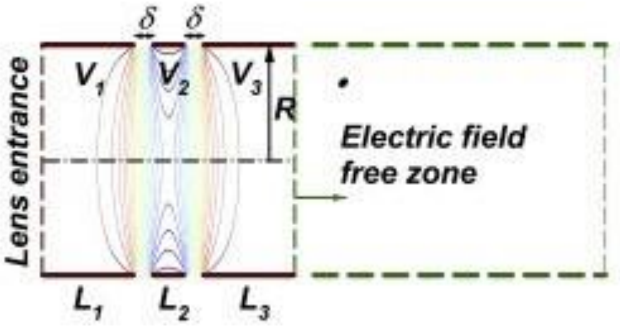


Figure 1: Einzel lens

Aberrations in ion optics: There Are two types of aberrations here Spherical Aberration - convergingof ions farther rom axis cross at different points than those near axis center (filed strength). Chromatic Aberration - Ion with different kinetic energies have different focal points we are mostly focusing on spherical aberration. The solution could be involving is to:

1. Skim off (Cut down) the outer part of the beam thereby losing ions and making the beam narrower.

2. Keep the lens diameter of the lens greater than ion beam size.

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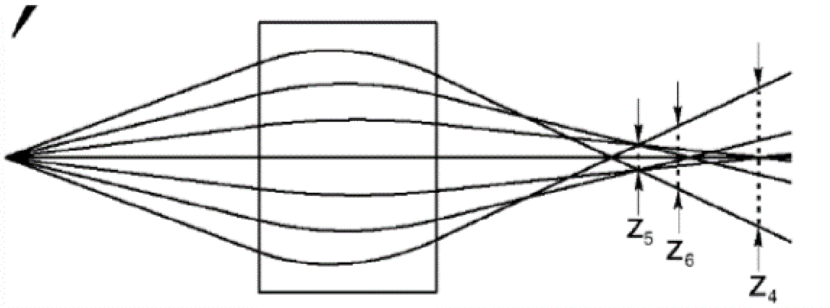


Spherical Aberration Fig 1

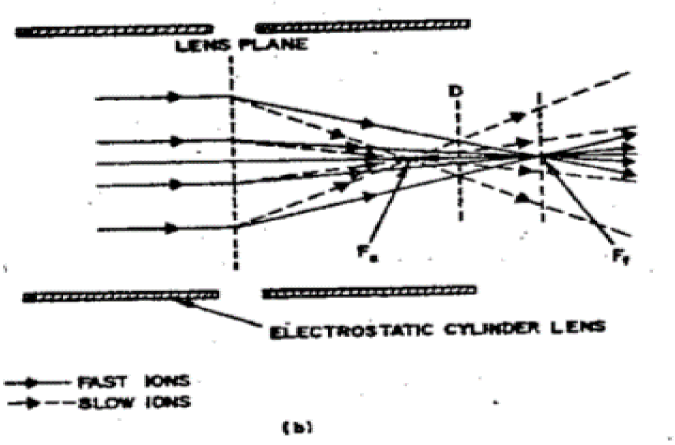


Fig 2

A tool which comes with such functionality reducing the experimental labor In order to achieve all of this we have used a SIMION the ion Simulation program discussed below :

SIMION

1. SIMION is an electrostatic lens analysis and design program.

2. In SIMION an electrostatic lens is defined as a two-dimensional electrostatic potential array containing both electrode and non-electrode points.

3. In addition, the user has the option of writing simple programs which can among other actions control field scale factors, dynamically adjust electrodes, and define explicit three-dimensional field functions (e.g. a quadrupole) used in lieu of array fields in specified portions of the potential array. Magnetic fields can be specified for computing ion trajectories in many electrostatic and magnetic field

4. In addition, the user has the option of writing simple programs which can among other actions control field scale factors, dynamically adjust electrodes, and define explicit three-dimensional field functions (e.g. a quadrupole) used in lieu of array fields in specified portions of the potential array. Magnetic fields can be specified for computing ion trajectories in many electrostatic and magnetic field environments.

5. SIMION provides a programming, visualization, and data recording environment for these simulations.

Working

1. We have first defined the physical geometry of the components in the system based on a uniform grid and sets the potentials (electric and magnetic) on the electrodes.

2. The program numerically solves the Laplace equation for each electrode

3. The electric and magnetic fields in the spaces between the electrodes, determines the potential at each grid point after the calculation.

4. The potential array is refined using over-relaxation methods allowing voltage contours and ion trajectories to be computed and plotted.

5. Planar and cylindrical symmetry assumptions allow the two-dimensional fields to support three-dimensional ion trajectory calculations.

Numerical Calculation in SIMION:

1. SIMION is designed to model the electrostatic fields and forces created by a collection of shaped electrodes given certain symmetry assumptions.
2. The electrostatic fields are modeled as boundary value problem solutions of a Laplace elliptical partial differential equation.
3. A finite difference technique called dynamically self-adjusting over-relaxation is applied to the two-dimensional potential array of points representing electrode and non-electrode regions to obtain a best estimate of the voltages for those points within the array that depict non-electrode regions.
4. A standard fourth-order Runge-Kutta method is used for numerical integration of the ion trajectory in two dimensions.

Simulation of General Einzel lens:

The Einzel has been simulated with a number of particles 10 with having charge -1 and maximum kinetic energy of 200 Kev and central electrode potential of 150 volts with first and second electrode potential that is V1 of 0 volts as those electrodes are having near ground potential with 10 potential contours as shown in Figure2

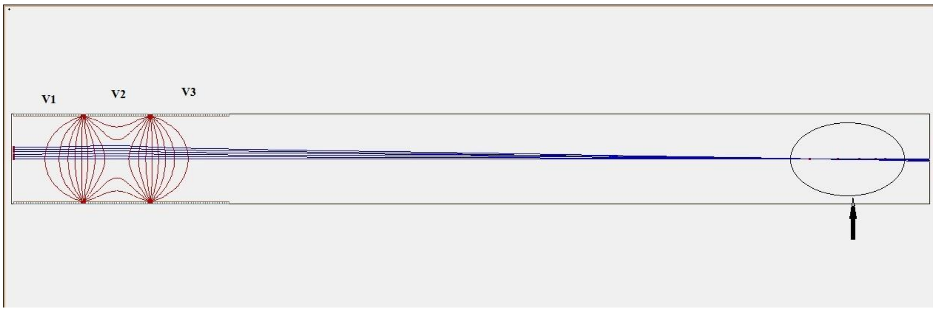


Figure 2: Einzel lens with particles flown

The nature of Aberration The ion optical nature of aberration of particles were flown can be shown in Figure3

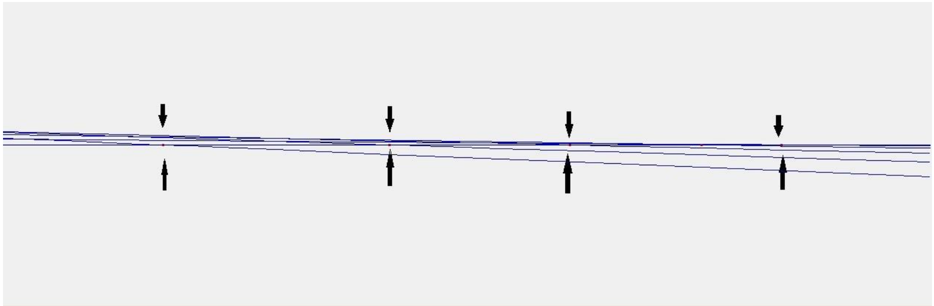


Figure 3: Aberration in Einzel lens

Aberration Calculation: The aberration can be calculated from the data recording function in SIMION in a textual form of tables later whose mean has been calculated of recorded x position of particle with settings as shown in Figure4

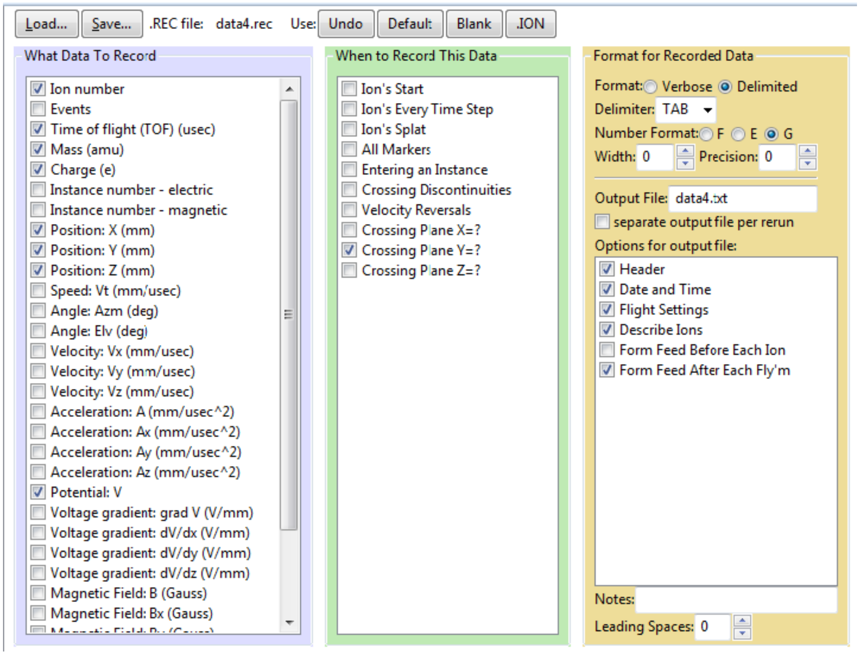
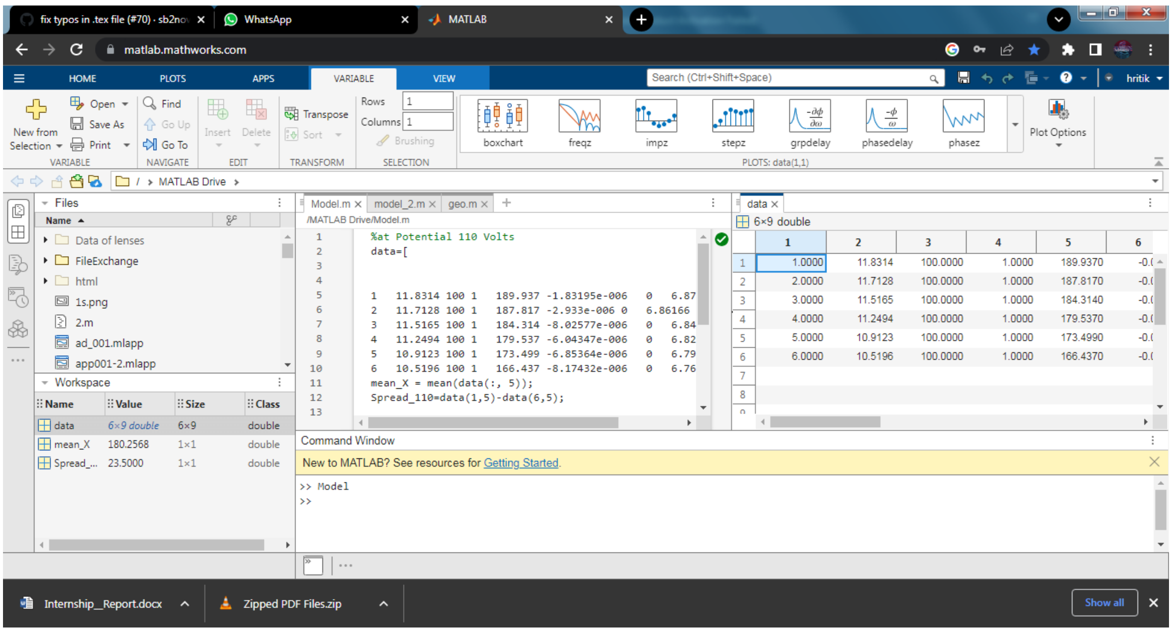


Figure 4: Data recording settings

MATLAB script: The calculation for the aberration has been done in MATLAB after exporting the data in textual format from simion



Nature of Aberration:

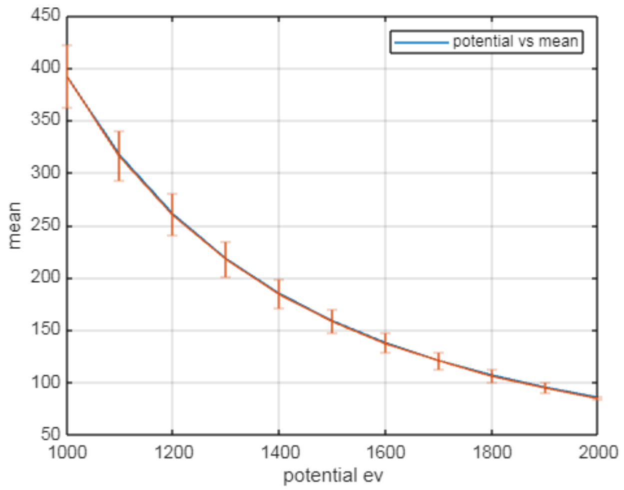


Figure 5: Aberration against Potential

For potentials ranging from 1000 ev to 2000 ev same data can be collected at each potential and at the interval of 100 ev and if we calculate the mean with error we can get the aberration plotted against potential with error would look like Figure5

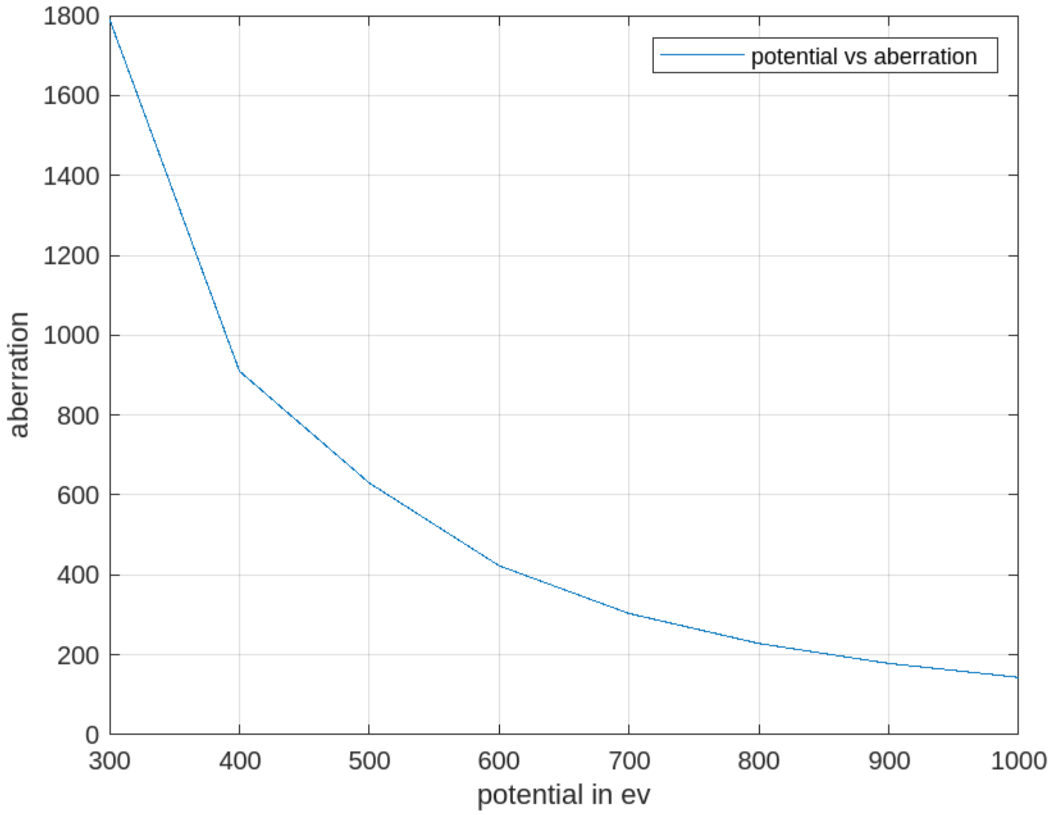
Data Collection: Once the geometries has been constructed using either LUA code or using a GEM file code and changing the coordinates to get various shapes of an Einzel lens. Here in this case the radiusto distance of electrode ratio has been changed to obtain various shapes of electrodes.

Field potential: After making a geometry, a lua code has been used to get the field at each of the points which is a big text file which tells us about the field and potential at each points the grid

Geometry: A MATLAB code here has been used to extract geometry from the raw data of the potentials. After getting the field data at each points of the grid the geometry has been obtained by subtractingthe calculated difference between main data (the raw potential data geometry) and the potential vector constructed.

0.4 Results

Nature of Aberration



For potentials ranging from 1000 ev to 2000 ev same data can be collected at each potential and at the interval of 100 ev and if we calculate the mean we can get the aberration plotted against potential

The nature of the aberration is such that at higher potentials there is a logarithmic decreasing

2) Here in this total of 9 geometries were created using the various radius by width ratio of Einzel lens andpotential has been collected for the same shown below:

Geometry 1 : r/d = default

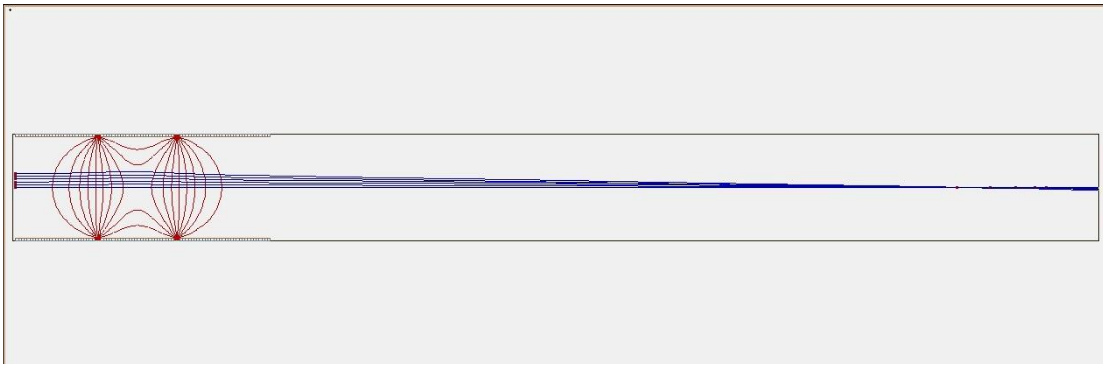


Figure 6: G1

Geometry 2 : r/d = 1/2

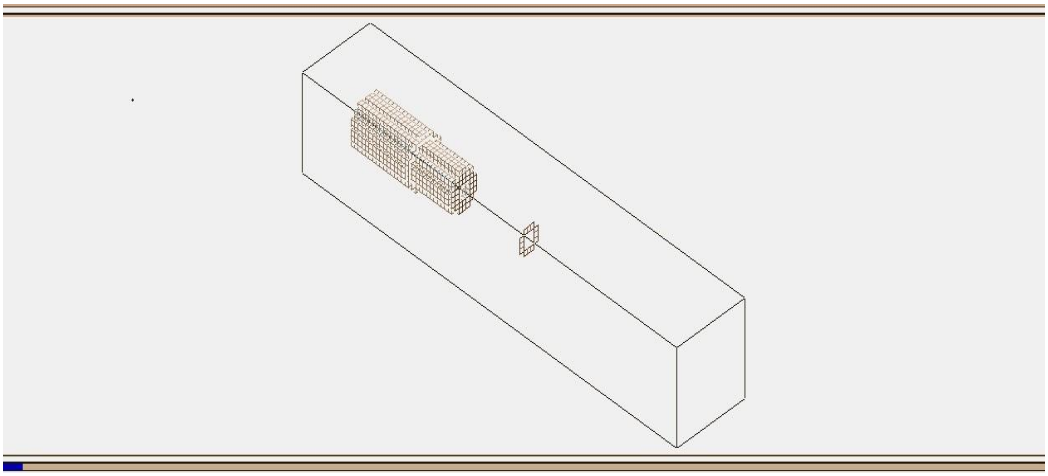


Figure 7: G2

Geometry 3 : r/d = 1

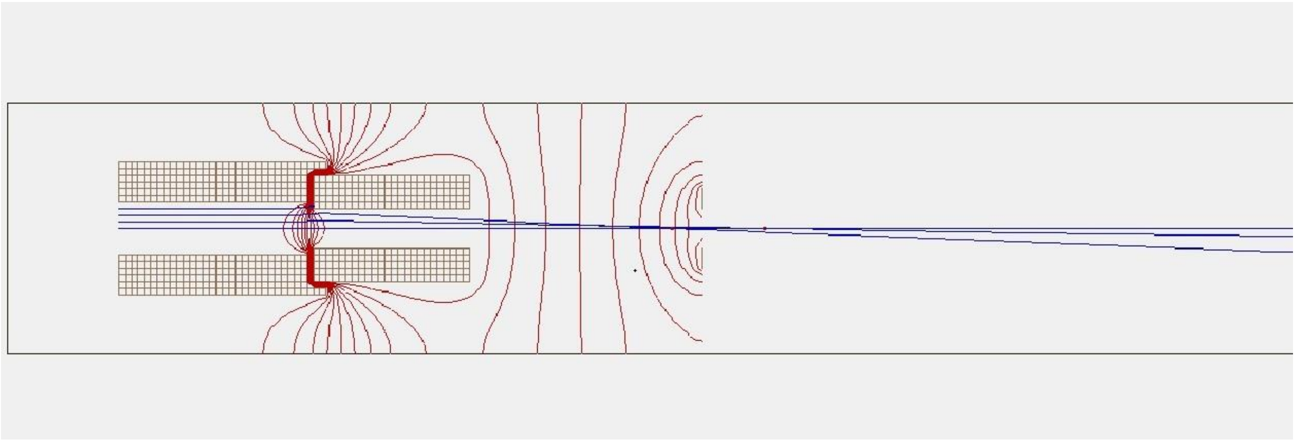


Figure 8: G3

Geometry 4 : r/d = 1/ 4

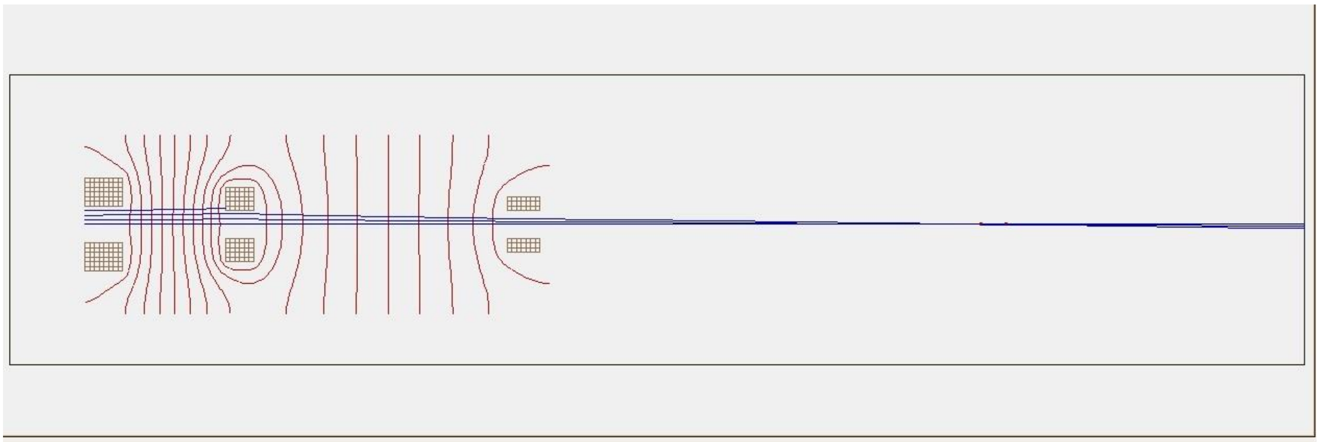


Figure 9: G4

Geometry 5 : r/d = 1/3

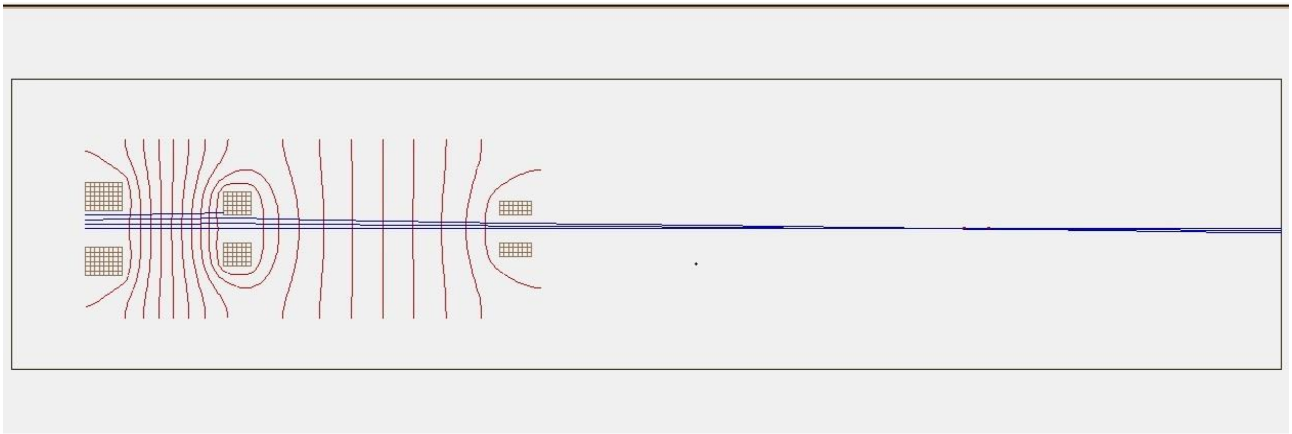


Figure 10: G5

Geometry 6 : r/d = 1/4

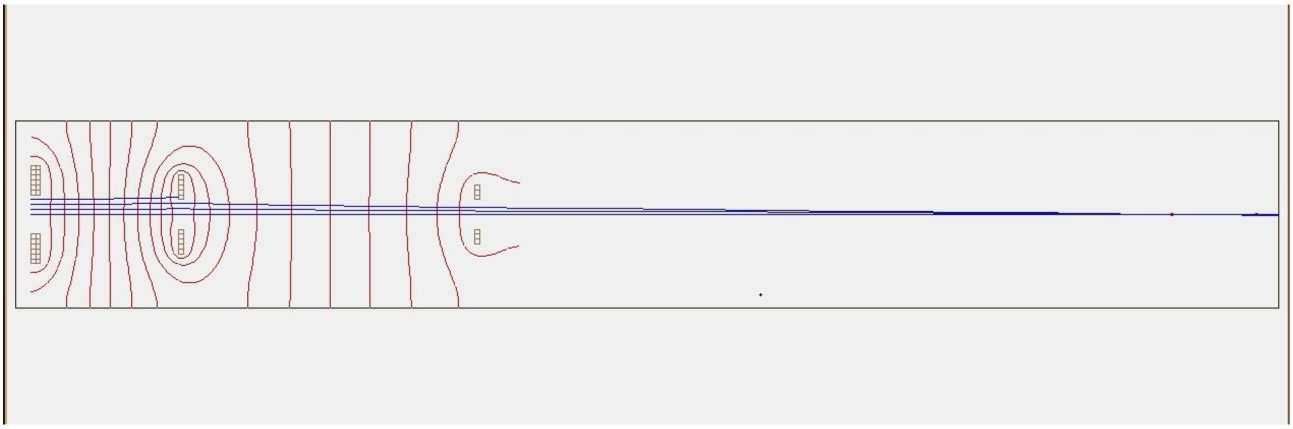


Figure 11: G6

Geometry 7 : r/d = 1/10

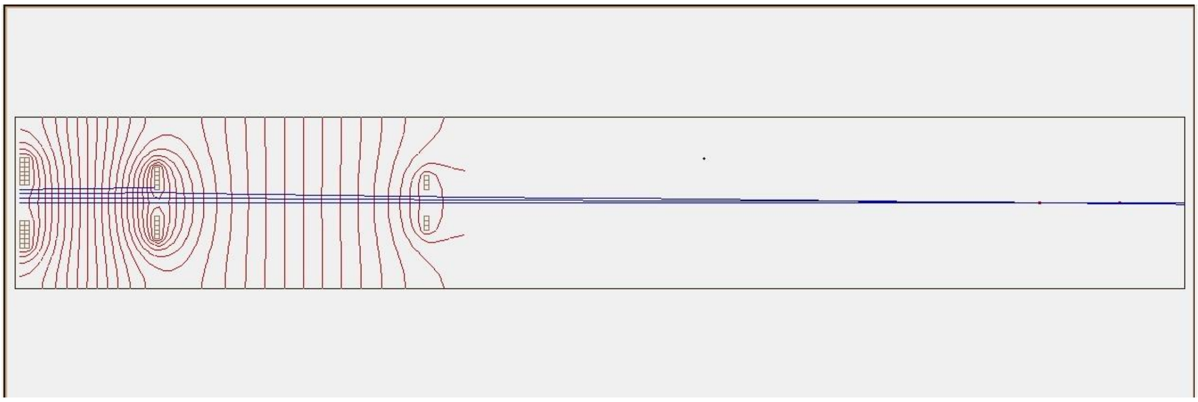


Figure 12: G7

Geometry 8 : r/d = 10

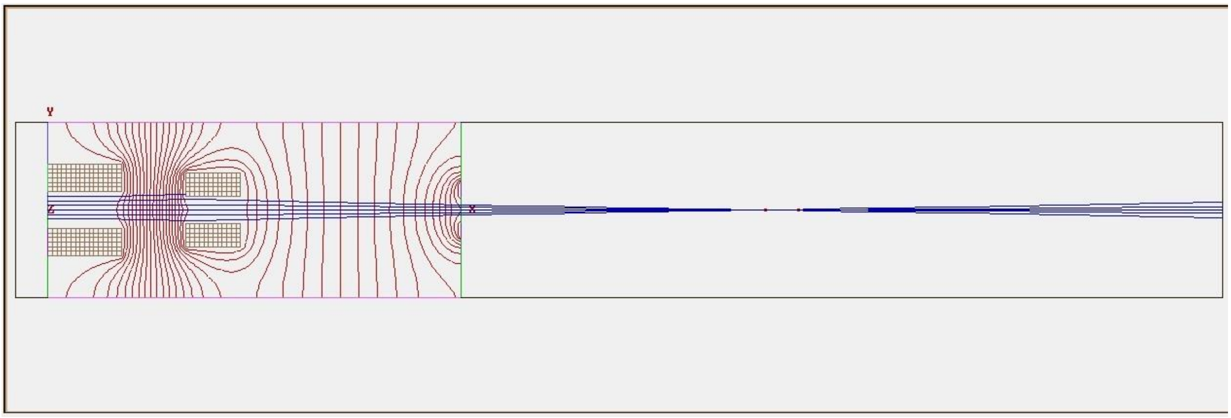


Figure 13: G8

Geometry 9: variation of spiral

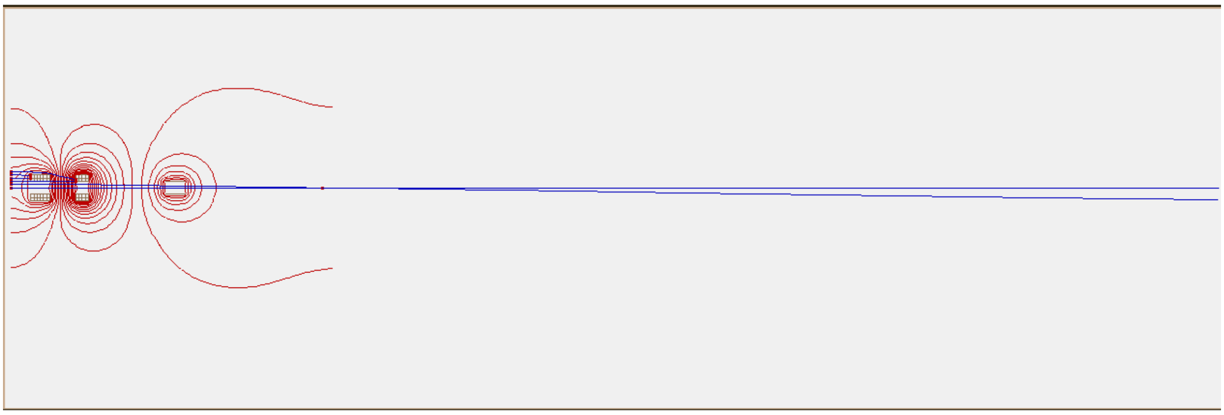


Fig 14:G9

0.5 Conclusions

1. Construction and Simulation of electrode geometries in SIMION using code various ways has been available in the software itself

2. The potential and field calculations for each point has been collected.

3. Optical parameters like Spherical Aberration, Magnification can be obtained.

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0.6 Future Directions

1. Different Geometries can be created other than regular Einzel lens by varying the coordinates such as a Conical shape or a Hexagon shape variations.

2. The data obtained from geometries like Aberration , Potentials and Geometries can be feed to an ML model

3. Supervised Learning will be applied to train a Machine learning model.

4. Developing the final Model by refining and iterating number of models for predicting optimal spherical Aberration constants and other optical parameters.

0.7 References

1) SIMION SUPPLEMENTAL DOCUMENTATION

2) SIMION COURSE MATERIALSIMION FOR PC

3) SIMION ADVANCED CLASS

4) ION OPTICS

5) MS short course 2011

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0.8 Acknowledgement

I would hereby like to thank the following persons for helping me in the above work under the guidance of: **Dr Abhishek Shahi** Department of chemistry GITAM School Of Sciences I am thankful for his expertise and feedback which greatly contributed to my growth in understanding of Ion Optics and Enlightening on research areas like Ion trapping, Electrostatic double storage rings, the computational sources and tools.

I also appreciate the sources and facilities provided by GITAM School of Sciences, Bengaluru.

Sincerely,

Ugale Hritik Ganesh